

5-4-1: Examining Firewalls and IPSs

At the end of this episode, I will be able to:

1. Identify and explain the fundamental role and functionality of firewall and intrusion detection and prevention systems, given a scenario.

Learner Objective: *Identify and explain the fundamental role and functionality of firewalls and intrusion detection and prevention systems, given a scenario.*

Description: In this episode, the learner will examine network firewalls, intrusion detection systems (IDS), and intrusion prevention systems (IPS). We will explore various types of firewalls and functionalities.

- What are network firewalls?
 - o Network Firewalls
 - ♣ These devices act as a security barrier between internal and external networks, controlling incoming and outgoing network traffic (north-to-south traffic flows) based on predetermined security rules. The primary purpose is to prevent unauthorized access while permitting outward communication.
- What are a few different types of firewalls and their associated functionality?
 - o Types of firewalls
 - ♣ Packet-filtering
 - ♣ Examines packets entering or leaving the network, either accepting or rejecting the packet based on user-defined rules.
 - ♣ Low level operation of the TCP/IP protocol stack, not examining the data in the packet.
 - ♣ Stateful packet inspection
 - ♣ Monitors the state of active connections, making decisions based on the context of the traffic and state of the network.
 - ♣ Provides a balance between performance and security.
 - ♣ Application-level
 - ♣ Filters incoming traffic between the network and the traffic source, operating at the application layer.
 - ♣ Inspects the data being sent and blocks harmful content or software.
 - ♣ Circuit-level
 - ♣ Circuit-level gateways or circuit-level firewall
 - ♣ Monitors TCP handshakes across the firewall to determine if the session is legitimate.
 - ♣ Operates at the session layer of the OSI model.
- Is there a firewall that bundles all of these various features?
 - o Next-Generation Firewalls (NGFW)
 - ♣ Incorporates features of traditional firewalls with quality of service (QoS) functionalities.
 - ♣ Includes additional features like application awareness, integrated intrusion prevention, and cloud-delivered threat intelligence.
 - o Intrusion Detection Systems (IDS)
 - ♣ Monitors networks or systems for malicious activity or policy violations.
 - ♣ Can be network-based (NIDS) or host-based (HIDS).
 - o Intrusion Prevention Systems (IPS)
 - ♣ Similar to IDS but with the ability to prevent detected threats.
 - ♣ Actively blocks or prevents intrusions in real-time.
 - o

Unified Threat Management (UTM)

- ♣ Combines and integrates various security services and features
 - ♣ Firewall
 - ♣ IDS/IPS
 - ♣ Antivirus
 - ♣ Gateway anti-spam
 - ♣ Content filtering
-

- Additional Resources:

- o Differences between stateful and circuit-level

- ♣ Stateful firewalls
 - ♣ Keeps track of the state of active connections, making decisions based on the context of the traffic and the state of the network.
 - ♣ Operate by inspecting both the header information and the contents of data packets.
 - ♣ Stateful inspections monitor ongoing connections to ensure that the traffic is associated with a known connection.
 - ♣ More of an awareness of the communications occurring over a network, can offer a higher level of security.
 - ♣ Remembers previous communications and makes security decisions based on past packets and the state of connections.
 - ♣ Circuit-level gateways:
 - ♣ Operate at layer 5 of the OSI model, making security decisions based on the handshaking protocols.
 - ♣ Monitor TCP handshaking between packets, determining whether a session request is legitimate.
 - ♣ These gateways can hide information about the p